

# Retrieval-Conditional Parsing Score (RCPS): Choosing Document Parsers by Retrieval, Not by Appearance

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## Abstract

Document RAG retrieves from parser outputs, yet parsers are commonly selected by intrinsic measures of transcription fidelity and boundary quality rather than by retrieval performance. We introduce the Retrieval-Conditional Parsing Score (RCPS), a training-free protocol that ranks parser-chunker combinations on a fixed held-out Q-A retrieval probe. RCPS averages standard MRR across specified retrievers and retrieval depths, using the same format-normalised exact-span relevance rule for every candidate. On KoGovDoc-RAG, table-enabled MinerU-on has a higher Boundary Clarity score than our production parser, Prod (0.713 versus 0.610). Yet its Hit@1 is 42.6 percentage points lower (0.123 versus 0.549). Across five parsers evaluated on all 294 pages, RCPS ranges from 0.137 to 0.584. With Prod fixed, four chunkers span only 0.535 to 0.593. A retriever-free coverage diagnostic finds no normalised exact match for 20.2% of reference spans in Prod output. No tested chunker splits more than 2.3%. On source-aligned OHR-Bench data, semantic noise lowers retrieval while Boundary Clarity remains stable or changes non-monotonically. In a secondary parser-training study, two direct preference optimization checkpoints have Hit@5 estimates only 0.95 and 1.15 percentage points above Prod on an audited six-domain compatibility subset. The workflow therefore selects with RCPS, diagnoses with coverage, and trains only if needed. We release the 663-Q-A KoGovDoc-RAG probe and an RCPS reference implementation; Appendix H identifies artifacts that remain pending.

## 1 Introduction

Document RAG retrieves from parser output, yet parsers are commonly selected by tran-

scription fidelity or Boundary Clarity (BC) (Zhao et al., 2025a) rather than by retrieval performance. Parsing precedes chunking, embedding, and retrieval, so omitted or corrupted content can cap downstream performance. These intrinsic metrics describe parser output but do not measure whether it preserves retrievable answers.

We therefore organise deployment into three steps: select by retrieval with RCPS, diagnose parser- versus chunker-side loss with coverage, and consider parser training only afterward (Figure 1).

RCPS is deliberately simple: it wraps standard retrieval metrics rather than introducing a new similarity function. The contribution is the deployment finding and the selection-and-diagnosis workflow.

Our production results show why selection must follow retrieval. Table-enabled MinerU-on (Niu et al., 2025) has higher BC than Prod (0.713 versus 0.610). Yet its Hit@1 is 0.123 versus Prod’s 0.549, a 42.6-point gap. On source-aligned OHR-Bench data (Zhang et al., 2025a), semantic corruption likewise lowers retrieval while BC remains stable or non-monotonic.

Coverage then locates the loss. For Prod, 20.2% of reference spans have no normalised exact match before chunking, while no tested chunker splits more than 2.3%.

In a three-parser end-to-end check, the RCPS top choice, Prod, also has the highest answer accuracy (72.5% versus 23.8% and 20.5%). The lower pair reverses, so this check supports only the top choice.

Training is secondary. After selection and diagnosis, we separately test whether parser-side training with retrieval-oriented feedback improves Prod. Two direct preference-optimization checkpoints have Hit@5 esti-

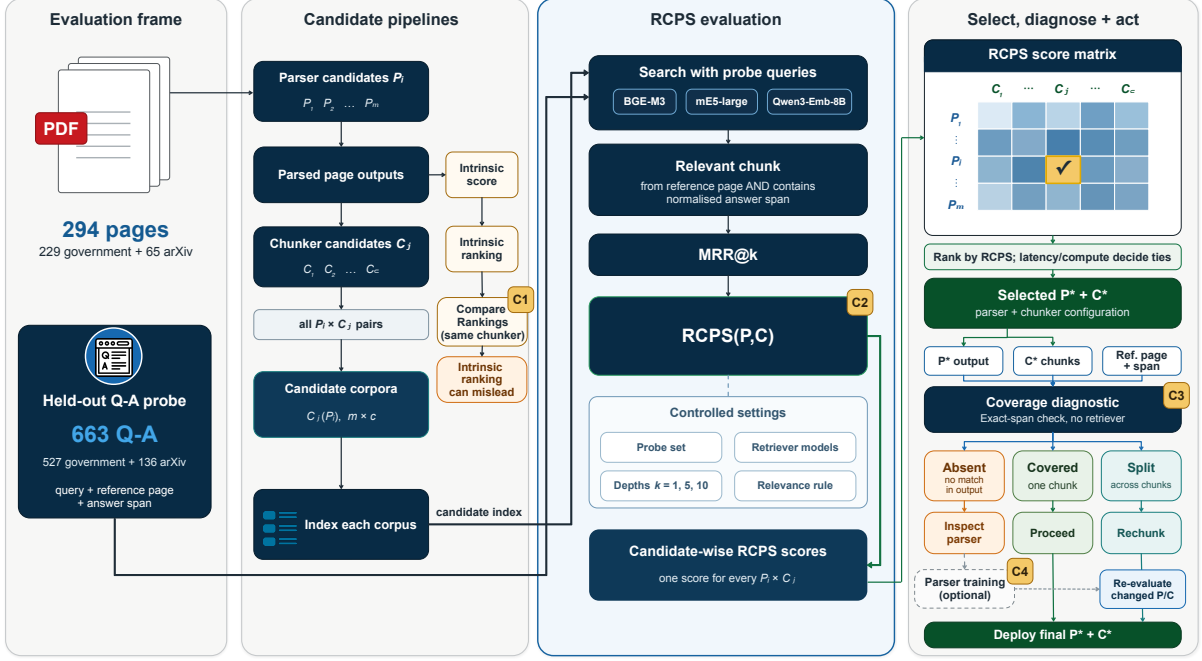


Figure 1: **RCPS workflow.** A held-out Q-A probe evaluates parser-chunker candidates. Intrinsic and RCPS parser rankings can disagree (C1); RCPS ranks candidates across fixed retrievers and retrieval depths (C2); coverage separates spans absent from parser output from those split across chunk boundaries (C3); and parser training is reserved for unresolved parser-side failures before changed configurations are re-evaluated (C4).

mates only about one point above Prod on an audited OHR-Bench compatibility subset. This frame differs from the selection comparison, so the effect sizes are not directly comparable.

**Contributions.** We show that intrinsic metrics can misrank parsers (C1). Our primary contribution is RCPS, a protocol for selecting parsers and chunkers by retrieval (C2). We then separate parser-side absence from chunker-side splitting (C3). Finally, we quantify the small, uncertain retrieval differences associated with parser training (C4). Our primary scope is answer-span retrievability; Appendix B reports the three-parser end-to-end check.

The repository contains the frozen 663-Q-A **KoGovDoc-RAG** probe and a reference implementation of RCPS.<sup>1</sup> The selection experiment indexed 294 pages, but the portable source-page mapping and some rerun artifacts are not yet packaged (Appendix H).

## 2 Related Work

**Evidence and gap.** Existing benchmarks document the parsing-retrieval mismatch. We turn this mismatch into a selection-and-diagnosis workflow. OHR-Bench (Zhang et al., 2025a) traces OCR noise through RAG. EnterpriseDocBench (Singh and Raj, 2026) and concurrent studies (Sun et al., 2026; Song, 2026) likewise find weak associations between parsing quality and retrieval. Most parsing benchmarks still rank intrinsic fidelity (Ouyang et al., 2025), while vision-based retrieval (Faysse et al., 2025) bypasses text parsing. We are unaware of prior work that combines a reusable parser-selection protocol with a diagnostic that separates absent from boundary-split spans.

**Training.** Retrieval-oriented training usually targets chunkers (Günther et al., 2024; Duarte et al., 2024; Zhao et al., 2024, 2025a; Singh et al., 2025), embeddings (Conti et al., 2025; Zhao et al., 2025b), retrievers, or readers (Nguyen et al., 2024; Chia et al., 2025; Yan et al., 2025). Our secondary study instead trains the parser. Prior parser training uses

<sup>1</sup><https://github.com/wigtn/WigtnOCR-RADP>

edit-distance and layout rewards (Wang et al., 2025) or query guidance (Wang et al., 2026). We test a retrieval reward and design a fidelity control. However, aligned per-Q-A results for the control are unavailable, so C4 makes no matched objective comparison. C1–C3 require no training.

### 3 Two Tools: Select and Diagnose

In our pipeline, a parser converts each page image into Markdown, a chunker divides that transcription into retrievable units, and a retriever ranks those units for a query. Appendix C specifies the parser input and output in detail.

RCPS and coverage answer different deployment questions. RCPS asks which parser-chunker combination retrieves best. Coverage asks whether an exact span is absent from parser output or divided during chunking.

#### 3.1 RCPS Selects by Retrieval

RCPS compares candidates on the same held-out Q-A probe. It wraps standard truncated mean reciprocal rank (MRR) rather than defining a new similarity function. For each query,  $\text{MRR}@k$  assigns  $1/j$  when the first relevant chunk appears at rank  $j \leq k$ , and zero when no relevant chunk appears in the top  $k$ ; the metric then averages these values over queries. Each candidate is evaluated on its own parsed-and-chunked corpus. Scores average specified retrievers, and every candidate uses the same *format-normalised exact-span relevance* rule. A chunk is relevant only if it comes from the reference page and contains the reference answer after shared whitespace and Markdown normalisation. We call Equation 1 the RCPS *score* and the complete procedure the RCPS *protocol*.

Given a parser  $P$ , a chunker  $C$ , a Q-A set  $D = \{(q_i, a_i, \text{page}_i)\}$ , a set of retrievers  $R$ , and retrieval depths  $K$ , RCPS averages MRR over all retriever-depth pairs:

$$\text{RCPS}(P, C) = \frac{1}{|R||K|} \sum_{r \in R} \sum_{k \in K} \text{MRR}@k(r, C(P), D). \quad (1)$$

Here,  $C(P)$  denotes the corpus obtained by parsing with  $P$  and then chunking with  $C$ .

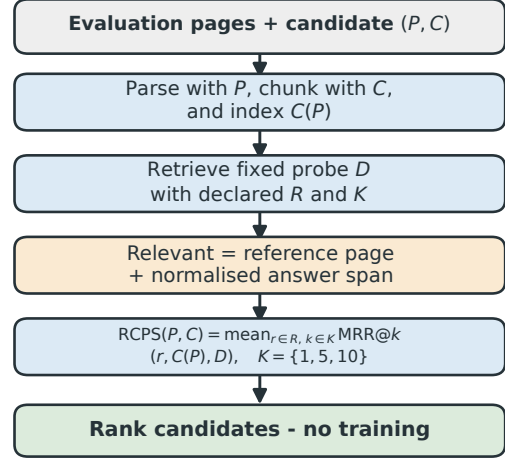


Figure 2: RCPS compares candidates on the same held-out Q-A probe. It averages standard MRR over declared retrievers and retrieval depths; a hit must come from the reference page and contain the normalised reference answer. Use a singleton retriever set when deployment is fixed, or a declared set when it is undecided. Evaluation itself requires no training.

We instantiate  $R$  with BGE-M3 (Chen et al., 2024), multilingual-e5-large (Wang et al., 2024), and Qwen3-Embedding-8B (Zhang et al., 2025b). We use  $K = \{1, 5, 10\}$ . Averaging over retrieval depths avoids tying selection to one depth. Figure 2 shows the workflow.

**Execution.** Build a representative held-out probe whose answers occur verbatim in their reference pages. Our probe is model-generated. An LLM-based check accepted 94/100 sampled pairs (§4.1). Parse each page once per parser, then chunk, index, and search every candidate corpus. For  $m$  parsers,  $c$  chunkers, and  $|R|$  retrievers, this requires  $m$  parsing runs and  $mc|R|$  retrieval evaluations.

**Interpretation.** The protocol needs neither training nor manual chunk labels because relevance follows from the answer span and source page. Fixing  $C$  ranks parsers; fixing  $P$  ranks chunkers. The ranking is relative to the candidate pool and probe.

#### 3.2 Coverage Separates Absence from Splitting

Coverage separates spans absent from parser output from spans split across chunk boundaries. It does not score retrievers. With parser output fixed, an answer is *covered* if its nor-

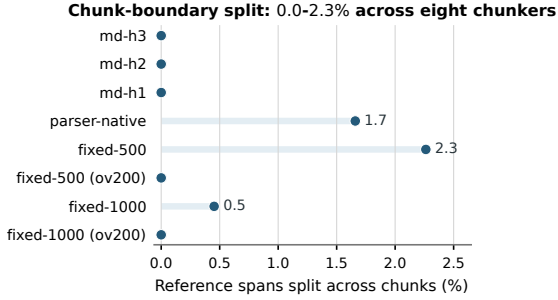


Figure 3: Coverage with Prod fixed (294 pages, 663 Q-A; no retriever). Absence is measured before chunking and stays 20.2%; changing the chunker affects only splitting, which reaches at most 2.3%.

malised span appears within one chunk. It is *split* if the span appears in the page output but crosses chunk boundaries. It is *absent* if the page output contains no exact match. Overlap may repair a split, whereas rechunking alone cannot restore an absent exact span. These labels are operational: *absent* does not prove semantic omission.

## 4 Experiments

### 4.1 Evaluation Frames and Setup

**Evaluation frames.** We report five frames that should not be compared directly. KoGovDoc-RAG uses frames of 294, 242, and 73 pages. The selection analysis retrieves all 663 Q-A against 294 pages: 229 government and 65 arXiv pages. Their evidence occurs on 242 pages; the other 52 pages are Q-A-free distractors. The training and mechanism analyses index the 242 evidence-bearing pages, while the pilot uses 73 pages and 202 Q-A. OHR-Bench (Zhang et al., 2025a) provides two further frames: 1,043 aligned Law-Manual Q-A for perturbations and a separate 2,036-Q-A, six-domain compatibility subset for training evaluation (Appendix A).

**Data and leakage control.** Qwen3-VL-30B produced pseudo-reference Markdown, which we manually de-noised. gpt-5.4-2026-03-05 then generated the Q-A. An LLM-based check accepted 94/100 sampled pairs. **Prod**, a Qwen3-VL-2B parser (Bai et al., 2025) fine-tuned for Korean documents, is the training baseline.

For MinerU, *MinerU-off* denotes the submitted output without table recognition, while *MinerU-on* denotes the later audited, table-

enabled output. The deployment comparison and its four-parser BC correlation use MinerU-on. MinerU-off is retained only as a separately labelled submitted-output diagnostic. Because the two runs also differ in software and retrieval environments, their differences do not isolate table recognition.

### 4.2 Boundary Clarity Can Misrank Parsers in This Pool (C1)

Boundary Clarity does not track RCPS in our candidate pool: outputs with the clearest boundaries can rank near the bottom on retrieval. Across five complete 294-page deployment configurations, RCPS spans 0.137–0.584. Vision-language parsers rank above OCR systems (Figure 4; Table 1). The 30B teacher and Prod lead at 0.584 and 0.583. MinerU-on has the highest measured BC among the complete-output parsers (0.713) but the lowest RCPS (0.137). PaddleOCR lacks measurable BC. Marker’s 0.073 score covers only 38 pages and is excluded from complete-output comparisons.

Among four 294-page parsers with defined BC, the Pearson correlation between BC and RCPS is  $r = -0.74$ . Adding Marker’s 38-page result gives  $r = -0.83$  ( $n = 5$ ). MinerU-on and Marker have the highest BC values ( $\approx 0.71$ ) but the lowest retrieval ranks. Both small-sample correlations are descriptive.

**Law-Manual perturbations (1,043 Q-A).** Within this aligned subset, semantic noise always lowers RCPS, whereas BC is stable, non-monotonic, or increasing. The study contains three benchmark outputs and twelve perturbations, not fifteen independent parsers (Appendix A). From mild to severe noise, MinerU RCPS falls  $0.476 \rightarrow 0.265$  while BC varies within 0.628–0.651. Qwen2.5-VL RCPS falls  $0.534 \rightarrow 0.497$  while BC stays near 0.563. GOT RCPS falls  $0.461 \rightarrow 0.298$  as BC rises  $0.586 \rightarrow 0.624$ . We claim no broader domain generality.

### 4.3 RCPS Ranks Parsers and Chunkers (C2)

On the same 663-Q-A probe and 294-page index, the 30B teacher and Prod are descriptively near-tied (0.584 versus 0.583). With Prod fixed, four chunkers form a point-

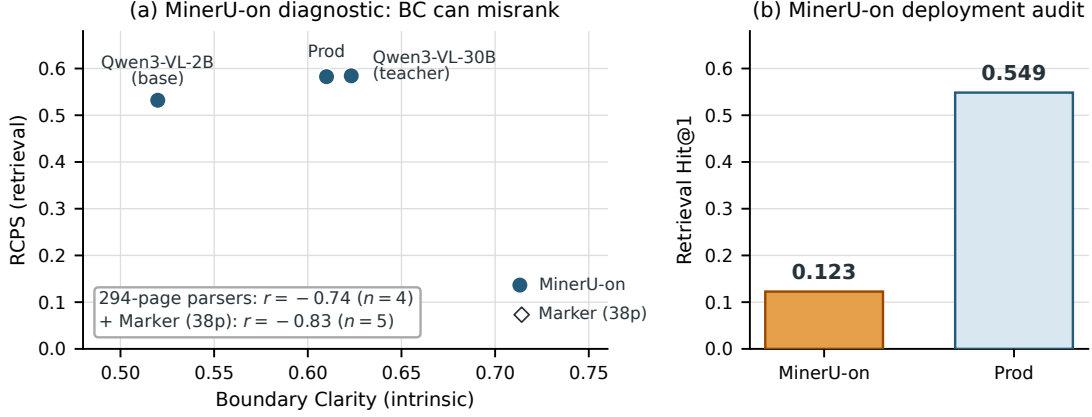


Figure 4: In this candidate pool, higher Boundary Clarity (BC) does not imply better retrieval. (a) Among four 294-page parsers with defined BC, the correlation with RCPS is  $r = -0.74$ ; adding Marker’s partial 38-page output gives  $r = -0.83$  ( $n = 5$ ). MinerU-on is used in this diagnostic. (b) Prod exceeds MinerU-on Hit@1 by 42.6 points (0.549 vs. 0.123; 4.47 $\times$ ).

estimate order from 0.593 to 0.535.

**Selecting parsers.** In Table 1, Prod reaches 0.583 RCPS and 0.549 Hit@1, versus MinerU-on’s 0.137 and 0.123. Prod’s Hit@1 is 4.47 $\times$  as high. The 30B teacher leads Prod by only 0.001 RCPS. Without paired uncertainty for this grid, we call them descriptively near-tied rather than equivalent. Latency and compute can break the deployment tie.

**Selecting chunkers.** With Prod fixed, the order is md-h3 (0.593) > parser-native (0.583) > LumberChunker (Duarte et al., 2024) (0.557) > fixed-500 (0.535). The chunker range is 0.058, versus 0.447 across the heterogeneous parser pool but 0.052 among only its three vision–language parsers.

| Parser                                   | BC           | CS   | RCPS         | Hit@1 |
|------------------------------------------|--------------|------|--------------|-------|
| <i>Deployment comparison (294 pages)</i> |              |      |              |       |
| Qwen3-VL-30B (teacher)                   | 0.623        | 3.38 | <b>0.584</b> | 0.545 |
| Prod (ours, 2B)                          | 0.610        | 3.07 | 0.583        | 0.549 |
| Qwen3-VL-2B (base)                       | 0.520        | 3.74 | 0.532        | 0.500 |
| PaddleOCR                                | —            | 3.46 | 0.140        | 0.125 |
| MinerU-on                                | <b>0.713</b> | —    | 0.137        | 0.123 |
| <i>Partial output</i>                    |              |      |              |       |
| Marker (38p)                             | 0.717        | 3.41 | 0.073        | 0.068 |

Table 1: RCPS ranks the 30B teacher and Prod at the top of the complete-output parser pool. Scores average three retrievers and  $k \in \{1, 5, 10\}$ . BC is Boundary Clarity (higher is better); CS is Chunk Stickiness (lower is better). The four-parser deployment correlation uses MinerU-on; Marker is excluded from complete-output comparisons.

**Retriever and depth robustness.** Use the deployment retriever when it is fixed. Otherwise, averaging hedges against an undecided retriever or tied candidates. An aggregate audit finds that a three-retriever MRR@10 average preserves the stored five-parser and four-chunker orders ( $\tau = 1.0$ ). Missing aligned per-Q–A arrays prevent a probe-resampling test. Appendix H reports the available single-retriever ablation and its configuration limits.

#### 4.4 Coverage Finds Pre-Chunking Absence (C3)

For Prod, 134/663 reference spans (20.2%) have no normalised exact match before chunking. No tested chunker splits more than 15/663 (2.3%) (Figure 3). This retriever-free diagnostic uses all 294 pages. Rechunking therefore cannot restore the exact match for roughly one answer in five.

MinerU-on’s absent rate is 66.1%, a 45.9-point gap over Prod under the same normalised exact matcher. Appendix C tests coverage-label robustness with alternative matchers, a cross-family judge, and human verification.

#### 4.5 Parser Training Is a Secondary, Bounded Lever (C4)

We call the optional parser-training stage after RCPS selection and coverage diagnosis Retrieval-Aware Document Parsing (RADP). RADP-aux aligns answer-span hidden states with frozen BGE-M3 embeddings through



an auxiliary contrastive loss. RADP-DPO samples multiple parses per page and ranks them with page-local BGE-M3 MRR. It converts higher- and lower-scoring parses into DPO preference pairs (Rafailov et al., 2023). RADP-Distill is a non-retrieval fidelity control that replaces the retrieval reward with edit distance to reference Markdown. Thus RCPS selects a pipeline without training, whereas RADP changes the parser after diagnosis.

All training data are page-disjoint from evaluation. Appendix E gives the common setup and preference construction. On the held-out 73-page pilot, neither RADP-aux nor RADP-DPO meets the target of a 5-point RCPS gain with a 95% CI lower bound above zero. On the audited 2,036-Q-A OHR-Bench compatibility subset, R2 and R3 exceed Prod Hit@5 by only 0.95 and 1.15 points (95% CIs [+0.33, +1.54] and [+0.31, +2.05]). These post-audit estimates are neither a full v2 rerun nor a pre-specified confirmation. Appendix F reports the full results, the reference-free SimPO control (Meng et al., 2024), and the unavailable aligned RADP-Distill comparison.

## 5 Discussion and Conclusion

**Select, diagnose, then train.** Parser choice changes Hit@1 by 42.6 points in the audited Prod-MinerU-on comparison. RADP-DPO is only about one Hit@5 point above Prod on a different OHR-Bench frame. These magnitudes are not directly comparable, but they make selection the first decision. In a three-parser end-to-end check, the RCPS top choice, Prod, also leads answer accuracy (72.5% versus 23.8% and 20.5%). The lower pair reverses, so this supports only the top choice.

**Clear boundaries do not guarantee retrieval.** BC measures boundary discontinuity, TextNED measures transcription similarity, and coverage tests exact-span preservation. MinerU-on combines high BC (0.713) with 66.1% exact-span absence and low Hit@1 (0.123), showing why these signals differ. DPO checkpoints have lower TextNED than Prod, but incomplete boundary evidence prevents a mechanism claim. Lower TextNED is therefore a post hoc association, not evidence that better fidelity caused retrieval gains (Ap-

pendix G).

**Practical recommendation.** Rank candidates with RCPS. Add overlap for *split* spans, inspect *absent* spans, and switch parsers when required content is missing. Train only afterward. Neither RADP method met the pilot target, and SimPO had negative point estimates. Retrieval should therefore select the parser. Coverage and fidelity diagnostics should guide what to improve next.

## Limitations

**Data scope and frames.** Denominators are not interchangeable. KoGovDoc-RAG selection indexes 294 pages for 663 Q-A. Training indexes the 242 evidence-bearing pages for the same Q-A, and the pilot uses 73 pages and 202 Q-A. The complete comparison covers five 294-page parsers. The descriptive BC correlations use the four with measurable BC ( $r = -0.74$ ) or add Marker on 38 pages ( $r = -0.83$ ). OHR-Bench uses 1,043 Law-Manual Q-A for perturbations and 2,036 Q-A for training compatibility. The former has three parser outputs plus twelve dependent perturbations and does not establish domain generality. We exclude mixed-version seven-domain artifacts (Appendix A).

**MinerU audit.** MinerU-off disables table recognition and is retained only as a submitted-output diagnostic. The separate MinerU-on audit reduces table-evidence absence from 87.9% to 41.7% (Prod: 13.9%). Its mixed-corpus RCPS is nevertheless 0.137, and its Hit@1 is 0.123 (Prod: 0.583 and 0.549). MinerU-on and MinerU-off have numerically similar BC (0.713 and 0.716), but their software and retrieval environments also differ. Changes in BC, retrieval, and absence therefore do not isolate table recognition. MinerU-on RCPS is 0.046 on government pages and 0.486 on arXiv pages.

**Reference, Q-A, and label quality.** Two 100-case checks answer different questions. An LLM check accepts 94/100 sampled Q-A, whereas a blind two-author study validates coverage labels. KoGovDoc-RAG uses manually de-noised Qwen3-VL-30B pseudo-references and LLM-generated Q-A. Neither complete set was human-verified. The fixed

probe controls query variation, not possible Qwen-lineage bias. OHR-Bench provides external checks, but the six-domain result is a compatibility analysis rather than a full v2 rerun. RCPS covers verbatim spans, not implicit or paraphrased answers. Coverage is matcher-defined. GPT-5.4 reclassifies 56% of Prod’s exact-match-absent cases as surface artefacts. The same GPT-5.4 checkpoint generates and judges end-to-end answers, so that check supports only the top parser choice.

**Training.** KoGovDoc-RAG R2 is exploratory. Its +1.96-point Hit@5 estimate (95% CI  $[-1.06, +5.03]$ ) uses 242 pages and a configuration selected after inspecting several settings. On the separate 2,036-Q-A OHR compatibility subset, R2 and R3 are +0.95  $[+0.33, +1.54]$  and +1.15  $[+0.31, +2.05]$  points above Prod. This subset was defined after a version audit and is not a full v2 rerun. We neither adjust for multiple checkpoints and secondary metrics nor cluster bootstrap resamples by shared evidence page. These are post-audit estimates, not the original confirmatory analysis.

**Scope.** RCPS targets answer-span retrieval rather than generated answers. The end-to-end experiment checks only three parsers. Preference candidates come from Prod, so other pools may differ. We do not test token-level reinforcement learning, curricula, or chunk-level oracle distillation. Broader parsers and training of chunkers, embedders, and rerankers remain future work.

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**Appendix roadmap.** Appendices A–D support the intrinsic-metric, RCPS-selection, and coverage analyses (C1–C3). Appendices E–G document the secondary parser-training study (C4), and Appendix H records dataset and release scope.

## A OHR-Bench Alignment and Noise Sensitivity

We use two OHR-Bench evaluation frames because the initial seven-domain artifacts mixed benchmark versions. They paired Q–A from the legacy 4,330-page release with parser outputs from the current 8,561-page bundle. In 223 legacy Q–A labelled *notes*, the old evaluation mapped the domain to v2 *administration*. The corresponding evidence is instead under v2 *textbook*, so the lookup selected unrelated documents and stored zeros. One further legacy textbook page, referenced by five Q–A, is absent from v2. After excluding these 228 affected cases, we retain a source-aligned compatibility subset of 2,036 Q–A across six domains. This subset supports compatibility checks, not a full v2 rerun. We therefore discard the correlation computed from the mixed-version frame.

The Law–Manual files retain matching document and page identifiers across both releases. We use this 1,043-Q–A frame for the C1 perturbation study. Its 15 variants comprise three benchmark outputs (reference, MinerU, and Qwen2.5-VL), three formatting perturbations, and nine semantic perturbations. The semantic variants apply three noise severities to MinerU, Qwen2.5-VL, and GOT; no unperturbed GOT output is available. Across the 15 variants, the descriptive BC–RCPS correlation is  $r = -0.35$ . This value should not be interpreted as a correlation across 15 independent parsers because variants within a family share the same source output. Table 2 reports the semantic families on the same Law–Manual pages with the same three-retriever average.

| Family     | Metric | Clean | Mild  | Mod.  | Severe |
|------------|--------|-------|-------|-------|--------|
| MinerU     | BC     | 0.657 | 0.628 | 0.651 | 0.631  |
|            | RCPS   | 0.595 | 0.476 | 0.384 | 0.265  |
| Qwen2.5-VL | BC     | 0.563 | 0.563 | 0.563 | 0.564  |
|            | RCPS   | 0.545 | 0.534 | 0.523 | 0.497  |
| GOT        | BC     | —     | 0.586 | 0.620 | 0.624  |
|            | RCPS   | —     | 0.461 | 0.385 | 0.298  |

Table 2: Semantic noise lowers RCPS within every family, while BC is stable, non-monotonic, or increasing. Results use 1,043 source-aligned Law–Manual Q–A. Clean denotes unperturbed output; Mild, Mod., and Severe denote increasing semantic noise. GOT has no clean output.

## B Does the RCPS Top Choice Lead on Answering?

We ask whether the RCPS top choice among Prod, MinerU-on, and PaddleOCR also leads in answer generation over all 663 KoGovDoc-RAG Q–A. BGE-M3 first retrieves the top five chunks from each parser corpus. gpt-5.4-2026-03-05 then generates an answer from those chunks and judges it against the reference answer.

This generation pipeline is not identical to the main RCPS protocol. The RCPS values in Table 3 average three retrievers at three retrieval depths, whereas answer generation uses only BGE-M3 top-five retrieval. Each row nevertheless uses the same parser output for its RCPS and generation results. In particular, the MinerU row uses MinerU-on. MinerU-off scores 0.212 RCPS, but it has no corresponding answer-generation run and is therefore excluded from this comparison.

Prod achieves 72.5% answer accuracy, compared with 23.8% for MinerU-on and 20.5% for PaddleOCR. Thus, the RCPS top choice also leads in judged answer accuracy. The nearly tied lower pair reverses, so this experiment does not validate their relative order. Moreover, the same model checkpoint generates and judges the answers. We therefore treat the experiment only as an end-to-end check of the top choice, not as validation of the full parser ranking.

| Parser      | Ans. acc.   | EM          | Answered | RCPS  |
|-------------|-------------|-------------|----------|-------|
| Prod        | <b>72.5</b> | <b>49.8</b> | 87.9     | 0.583 |
| MinerU (on) | 23.8        | 14.5        | 39.1     | 0.137 |
| PaddleOCR   | 20.5        | 11.5        | 45.9     | 0.140 |

Table 3: The RCPS top choice, Prod, also has the highest answer accuracy in this three-parser check; the near-tied lower pair reverses. Generation uses BGE-M3 top-five retrieval over 663 Q–A, with MinerU-on used for both RCPS and generation. RCPS averages the main three-retriever grid at three retrieval depths. EM is normalised-span exact match; Answered excludes NO ANSWER. Ans. acc., EM, and Answered are percentages; RCPS uses a 0–1 scale. The same GPT-5.4 checkpoint generates and judges answers, so this checks only the top choice.

## C Is the Absent Label Robust?

This appendix assesses the exact-span absent label with alternative matchers, a cross-family judge, and a parser-masked human check.

### Parser I/O and the “absent” label.

Each PDF page is first rendered as an image. A document parser maps that image to a linear Markdown transcription. The intended output preserves body text in reading order, ATX (#-style) headings, and GitHub-flavoured Markdown tables. It discards layout coordinates, fonts, figure imagery, and page furniture unless they are transcribed as text. The input collection includes born-digital and scanned PDFs, multi-column layouts, merged table cells, stamps, seals, and text embedded in figures.

Only the resulting Markdown is chunked, embedded, and retrieved. Under the operational definition in §3.2, a reference answer is *absent* when its normalised exact span has no match in the source-page transcription. The reference Markdown used for TextNED and the reference answer spans come from manually de-noised Qwen3-VL-30B output. We therefore treat both as pseudo-ground truth.

**Worked example: a corrupted table unit.** On KoGov page val\_0155, the pseudo-reference and parser output differ as follows:

|                         |                      |
|-------------------------|----------------------|
| Pseudo-reference answer | $A = 180\text{ m}^2$ |
| MinerU-on output        | $A = 180\text{m}$    |

MinerU-on omits the exponent. The shared normalisation removes formatting, but it does not treat  $\text{m}^2$  and  $\text{m}$  as equivalent. The complete answer is therefore absent under the exact-span relevance rule. Rechunking cannot reconstruct the missing exponent.

**Alternative matchers.** The reference Markdown, reference answers, and Prod share Qwen3 lineage. To test whether this shared lineage explains the absent-rate gap, we apply five model-free presence matchers to all 663 Q–A. L0 uses raw substring matching. L1 applies Unicode Normalization Form KC (NFKC), case folding, whitespace normalisation, and Markdown normalisation. L2 additionally removes digit separators and in-content punctuation. L3 requires at least 90% order-free token recall. L4 requires the

longest common substring to cover at least 80% of the answer. L3 and L4 are alternative tolerant criteria, not successive steps in a nested ladder. If surface form explains most of the gap, at least one tolerant matcher should narrow it.

| Absent %       | L0    | L1    | L2    | L3    | L4    |
|----------------|-------|-------|-------|-------|-------|
| Prod (2B)      | 24.1  | 20.2  | 19.6  | 24.1  | 16.9  |
| 30B teacher    | 22.8  | 18.9  | 16.6  | 19.2  | 16.3  |
| 2B base        | 30.0  | 25.2  | 24.7  | 27.0  | 21.1  |
| MinerU-off     | 74.8  | 70.4  | 68.0  | 73.0  | 68.6  |
| PaddleOCR      | 67.0  | 62.7  | 62.0  | 59.9  | 60.5  |
| MinerU-Prod    | +50.7 | +50.2 | +48.4 | +48.9 | +51.7 |
| PaddleOCR-Prod | +42.8 | +42.5 | +42.4 | +35.7 | +43.6 |

Table 4: The OCR-parser absence gaps over Prod persist across five presence matchers (663 Q-A, no retriever; MinerU-off). L0 is raw matching; L1 is normalised exact matching; L2 removes additional punctuation; L3 uses token recall; and L4 uses longest-common-substring coverage. The two other Qwen3-derived parsers remain within 5.9 percentage points of Prod.

**Cross-family model judge.** We next ask GPT-5.4, a different model family from the evaluated parsers, whether evidence remains recoverable in each L1-absent parser-answer pair (Table 5). *Present* denotes a recoverable surface mismatch. *Degraded* means that evidence remains but is too corrupted for reliable retrieval, and *absent* denotes missing content. We combine the latter two labels as *retrieval-unusable*. The same procedure is applied independently to every parser; the MinerU analysis here uses MinerU-off.

The Q-A were also generated by a GPT-family model, so this judge is not independent of Q-A generation. The model-free matchers above avoid that dependence. The judge marks 56% of Prod’s L1-absent cases as surface artefacts, compared with about 16% for both MinerU and PaddleOCR. After those artefacts are removed, the MinerU-Prod retrieval-unusable gap remains 50.4 percentage points. The corresponding gaps are 50.2 points under L1 exact matching and 51.7 points under the L4 OCR-tolerant criterion.

**Human verification of absent labels.** Two authors independently labelled 100 parser-masked L1-absent cases, stratified by question type. The sample contains 50

|            | present<br>(artefact) | degraded | absent | retrieval-<br>unusable |
|------------|-----------------------|----------|--------|------------------------|
| Prod       | 56%                   | 2%       | 42%    | <b>8.9%</b>            |
| MinerU-off | 16%                   | 9%       | 75%    | <b>59.3%</b>           |
| PaddleOCR  | 16%                   | 22%      | 63%    | <b>52.8%</b>           |

Table 5: A cross-family judge preserves the OCR-parser gap after recoverable surface mismatches are separated. GPT-5.4 labels 1,017 parser-answer cases marked absent by L1 (134 Prod, 467 MinerU-off, and 416 PaddleOCR). Degraded and absent are retrieval-unusable; the first three columns are percentages within each parser’s L1-absent set, whereas the final column is a percentage of all 663 Q-A.

MinerU-on, 30 Prod, and 20 PaddleOCR cases. This study evaluates coverage labels; it is separate from the 100-pair Q-A quality check. Before adjudication, the authors agreed on 81/100 cases, with  $\kappa = 0.615$ . They then jointly resolved the remaining 19 cases.

After adjudication, 42/50 MinerU-on cases were retrieval-unusable (84.0%, Wilson 95% CI [71.5, 91.7]). The result was 12/30 for Prod (40.0%, [24.6, 57.7]) and 19/20 for PaddleOCR (95.0%, [76.4, 99.1]).

Ninety-three human-labelled cases also appear in the automated-judge set: 43 MinerU, 30 Prod, and 20 PaddleOCR cases. We exclude seven MinerU-on cases that have no corresponding MinerU-off instance. On the matched set, the judge’s retrieval-unusable rates are 84.2% for MinerU, 84.1% for PaddleOCR, and 44.0% for Prod. Pooled binary human-judge agreement is 84/93 (90.3%). The human labels thus confirm the direction of the OCR-parser gap in this stratified sample. Different sampling fractions and MinerU configurations preclude a population-level replication claim.

**Causes of absence.** For Prod, exact-match absent counts are 23/165 (13.9%) for table-evidence answers, 43/201 (21.4%) for factoid answers, and 63/290 (21.7%) for procedural answers. The count is 5/7 (71.4%) for figure-evidence answers, but this seven-case estimate should not be generalised. Its high point estimate is consistent with figure imagery not being transcribed into retrievable text.

We observe three recurring causes. First, parsers drop table cells before the content

reaches Markdown. For table-evidence answers, absent rates are 87.9% for MinerU-off, 41.7% for MinerU-on, and 13.9% for Prod. Second, parsers skip text embedded in captions, stamps, seals, or figure labels. Third, they misrecognise numerals or units beyond the L2/L4 tolerance.

## D Chunkers Rarely Split Exact Spans

We apply the retriever-free coverage diagnostic to Prod output for all 663 Q–A over 294 pages. As defined in §3.2, text matching labels each answer as covered, split, or absent. Because the parser output is fixed, the parser-defined absent rate remains 20.2% across all eight chunkers. No tested chunker splits more than 2.3% of exact answer spans.

| Chunker         | covered | split | absent |
|-----------------|---------|-------|--------|
| md_h3           | 79.8%   | 0.0%  | 20.2%  |
| md_h2           | 79.8%   | 0.0%  | 20.2%  |
| md_h1           | 79.8%   | 0.0%  | 20.2%  |
| parser_native   | 78.1%   | 1.7%  | 20.2%  |
| fixed500        | 77.5%   | 2.3%  | 20.2%  |
| fixed500_ov200  | 79.8%   | 0.0%  | 20.2%  |
| fixed1000       | 79.3%   | 0.5%  | 20.2%  |
| fixed1000_ov200 | 79.8%   | 0.0%  | 20.2%  |

Table 6: No tested chunker splits more than 2.3% of answers; the pre-chunking absent rate stays 20.2% (Prod, 294 pages, 663 Q–A). `parser_native` uses blank-line paragraph boundaries; `md_hk` uses Markdown heading depth  $k$ ; `fixed N[_ovM]` uses  $N$ -character windows with optional  $M$ -character overlap.

## E How RADP-DPO Builds Preferences (C4)

**Common training setup.** All variants are LoRA fine-tunes of Prod ( $r = 8$ ,  $\alpha = 32$ ) (Hu et al., 2022). Evaluation uses deterministic decoding with temperature 0 and a 1,536-token limit. Candidate generation for preference construction is stochastic. The 2,667-page training corpus, which supplies 6,164 generated Q–A, is page-disjoint from every evaluation frame. RADP-aux sweeps  $\lambda \in \{0, 0.1, 0.3, 0.5\}$  for its contrastive loss. Unless stated otherwise, confidence intervals use 1,000 paired Q–A-level percentile-bootstrap resamples.

**Preference construction.** RADP-DPO samples alternative Prod outputs for each

page in the separate training corpus. It chunks every candidate parse and scores those chunks against questions linked to the page, with other-page chunks as distractors. The page-local reward is BGE-M3 MRR averaged over  $k \in \{1, 5, 10\}$ . This training reward differs from the evaluation RCPS, which averages three retrievers. Multilingual-e5-large and Qwen3-Embedding-8B are therefore unseen by the reward.

Candidate parses that differ by at least 5 percentage points in page-local reward form a DPO pair. The higher-scoring parse is the chosen example. R1 samples  $K = 2$  parses with uniform distractors. R2 starts from the R1 checkpoint and trains on a second preference round. R3 expands the candidate pool to  $K = 16$  by adding 14 parses to the two R1 candidates. It also adds full-corpus hard negatives: chunks from other pages that are closest to the current queries.

**Training objective.** Let  $x$  be the input page, and let  $y^+$  and  $y^-$  be its chosen and rejected parses. Define  $\Delta_\theta = \log \pi_\theta(y^+ | x) - \log \pi_\theta(y^- | x)$  and  $\Delta_{\text{ref}} = \log \pi_{\text{ref}}(y^+ | x) - \log \pi_{\text{ref}}(y^- | x)$ . During training,  $\pi_\theta$  enables the trainable LoRA adapters, whereas  $\pi_{\text{ref}}$  disables them:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma(\beta[\Delta_\theta - \Delta_{\text{ref}}]). \quad (2)$$

The SimPO control uses a length-normalised chosen–rejected log-probability margin ( $\beta = 2.0$ , target margin  $\gamma = 1.0$ ) without a reference policy.

## F Parser-Training Results Across Frames (C4)

Across evaluation frames, the observed training differences are small or uncertain. The 73-page pilot misses its target. Every DPO and SimPO interval in the 242-page KoGovDoc-RAG analysis crosses zero, while the OHR-Bench compatibility estimates are about one Hit@5 point.

**Evaluation frames.** Before the pilot, success meant a gain of at least 5 RCPS points with a 95% CI lower bound above zero. The pilot held out 73 pages and 202 Q–A from 242 evidence-bearing pages. Its best RADP-aux

and RADP-DPO estimates missed that target, and their intervals crossed zero.

The later KoGovDoc-RAG analysis pools the 169 development pages with the 73 held-out pages. It is therefore not an independent holdout. All 663 Q–A have evidence on these 242 indexed pages; the 294-page selection index additionally contains 52 Q–A-free distractor pages. The cross-domain analysis instead uses the 2,036-Q–A, six-domain OHR-Bench compatibility subset in Table 7. All training examples come from the separate 2,667-page corpus, and every evaluated system is a LoRA fine-tune of Prod.

**KoGovDoc-RAG results.** The three RADP-DPO milestones have parser-native Hit@5 point estimates 1.96–2.11 percentage points above Prod. For each checkpoint, the paired-bootstrap proportion  $P_{\text{boot}}[\Delta\text{Hit@5} > 0]$  is about 0.90 (Table 8). This descriptive bootstrap proportion is not a  $p$ -value, and every two-sided interval crosses zero at  $n = 663$ .

These estimates are exploratory. The reporting configuration was selected after examining multiple chunker, retriever, and retrieval-depth settings, without a family-wise multiplicity correction. The bootstrap also re-samples Q–A pairs rather than evidence pages. Table 7 omits R1 because no source-aligned cross-domain array is available for that checkpoint. It also omits RADP-Distill because no aligned per-Q–A artifact is available on the same evaluation frame.

**RADP-aux does not meet the pre-specified criterion.** On the held-out 73-page, 202-Q–A fold,  $\lambda = 0.1$  gives the largest estimate relative to the  $\lambda = 0$  fine-tuning control: +1.13 RCPS percentage points with md-h3 and +2.31 with parser-native chunking. Estimates decline at larger values of  $\lambda$ , and every paired confidence interval relative to  $\lambda = 0$  includes zero.

**SimPO has negative point estimates.** Relative to Prod on the 242-page fold, its three-retriever mean Hit@5 difference is  $-0.85$  percentage points with md-h3 and  $-0.70$  with parser-native chunking; both confidence intervals cross zero. These results do not isolate whether reference removal, length normalisa-

tion, or another optimisation difference causes the change.

**Unavailable fidelity control.** RADP-Distill ranks candidates by edit distance rather than retrieval, but an aligned per-Q–A artifact is unavailable. We therefore omit its cross-retriever and evidence-type breakdowns and make no quantitative comparison between training objectives.

## G Do Training Point Estimates Track Fidelity or Boundaries? (C4)

The available measurements do not identify why the training point estimates change. DPO checkpoints have lower TextNED than Prod, but TextNED does not reproduce their retrieval order and the boundary evidence is incomplete. Table 9 reports Boundary Clarity (BC), normalised edit distance to reference Markdown (TextNED), and chunking shape on the 242-page corpus. TextNED is the character-level Levenshtein distance divided by the longer string length.

| Variant           | len  | c/pg | c/len | BC $\uparrow$ | TextNED $\downarrow$ |
|-------------------|------|------|-------|---------------|----------------------|
| Prod (baseline)   | 1385 | 4.82 | 274   | 0.630         | 0.240                |
| aux $\lambda=0.0$ | 799  | 2.96 | 245   | 0.553         | 0.626                |
| aux $\lambda=0.1$ | 1380 | 4.05 | 321   | 0.652         | 0.423                |
| aux $\lambda=0.3$ | 1930 | 6.73 | 272   | 0.470         | 0.332                |
| aux $\lambda=0.5$ | 2035 | 5.82 | 327   | 0.556         | 0.318                |
| DPO-R1            | 1606 | 4.97 | 311   | 0.646         | 0.167                |
| DPO-R2            | 1610 | 4.83 | 321   | 0.647         | <b>0.163</b>         |
| DPO-R3            | 1514 | 4.88 | 300   | —             | 0.185                |
| SimPO             | 1703 | 5.31 | 305   | 0.601         | 0.186                |

Table 9: DPO checkpoints have lower TextNED than Prod on the 242-page corpus, but the metric does not reproduce their retrieval order. Len is mean parsed characters per page, c/pg is mean chunks per page, and clen is mean characters per chunk. BC was not computed for R3.

**DPO checkpoints have lower TextNED.** The three RADP-DPO checkpoints have TextNED values of 0.163–0.185, compared with Prod’s 0.240. Relative to the pseudo-reference, these distances are 23–32% lower. RADP-aux instead has TextNED values of 0.318–0.626.

All three DPO checkpoints also have positive Hit@5 point estimates in Table 8, but TextNED does not order them. R2 has the lowest TextNED and the smallest DPO Hit@5 difference. R3 has the highest TextNED



| $\Delta$ vs Prod (pp) | Hit@1 | Hit@5 | Hit@10 | MRR@10 | nDCG@5 |
|-----------------------|-------|-------|--------|--------|--------|
| RADP-DPO-R2           | +0.59 | +0.95 | +0.90  | +0.78  | +0.82  |
| RADP-DPO-R3           | +1.46 | +1.15 | +0.90  | +1.30  | +1.28  |

Table 7: R2 and R3 have small positive deltas on the audited OHR-Bench compatibility subset, but this is not a full v2 rerun. The subset contains 2,036 source-aligned Q-A across six domains and uses a three-retriever average with 1,000 Q-A-level paired-bootstrap resamples. Deltas are percentage points; Hit@5 intervals are R2 [+0.33, +1.54] and R3 [+0.31, +2.05].

| Variant (vs Prod = 0.6757) | Hit@5  | $\Delta$ Hit@5 [95% CI] | $P_{\text{boot}}[\Delta\text{Hit@5} > 0]$ | $\Delta$ RCPS (pp) |
|----------------------------|--------|-------------------------|-------------------------------------------|--------------------|
| RADP-DPO-R3                | 0.6968 | +2.11 [−0.90, +5.13]    | 0.91                                      | +1.72              |
| RADP-DPO-R1                | 0.6963 | +2.06 [−0.96, +5.13]    | 0.91                                      | +0.57              |
| RADP-DPO-R2                | 0.6953 | +1.96 [−1.06, +5.03]    | 0.90                                      | +0.47              |
| RADP-SimPO (ctrl)          | 0.6687 | −0.70 [−3.77, +2.31]    | 0.32                                      | −1.56              |

Table 8: All KoGovDoc-RAG DPO point estimates are positive, but every two-sided interval crosses zero; the SimPO point estimate is negative. The analysis indexes 242 pages for 663 Q-A and uses parser-native chunking with 10,000 Q-A-level paired-bootstrap resamples. Hit@5 averages three retrievers; Prod is 0.6757.  $P_{\text{boot}}[\Delta\text{Hit@5} > 0]$  is a bootstrap proportion, not a  $p$ -value. Across three independent R1-recipe seeds, the standard deviation of mean  $\Delta\text{Hit@5}$  is 0.90 percentage points.

among the DPO checkpoints and the largest Hit@5 difference. The co-occurrence is consistent with fidelity as one possible explanation, but these observational results do not identify a causal mechanism.

**Available structure evidence is incomplete.** R1 and R2 have 4.97 and 4.83 chunks per page, close to Prod’s 4.82. Their mean chunk lengths, however, increase from 274 characters for Prod to 311 and 321. Their BC values are 0.646 and 0.647, compared with Prod’s 0.630. R3 has comparable chunking shape, but no reproduced BC value. The missing R3 value and the lack of uncertainty estimates prevent a systematic conclusion about boundary change.

**Interpretation.** One possible explanation for RADP-aux’s smaller point estimates is that its hidden-state objective can affect retrieval only indirectly through generated Markdown. Previously computed OHR TextNED and RADP-Distill breakdowns use artifacts that are not aligned with the audited subset, so we exclude them from the mechanism analysis.

## H Dataset Composition and Release Status

**Dataset composition.** KoGovDoc-RAG contains 229 Korean government pages and 65 arXiv pages, contributing 527 and 136

verbatim-answerable Q-A. The separate 2,667-page training set is page-disjoint from all 294 evaluation pages. Prod exceeds its 2B base by 4.9 Hit@1 points, whereas the audited Prod-MinerU-on difference is 42.6 points. This comparison provides scale, not a formal bound on fine-tuning effects.

**Stored retriever ablation.** This ablation predates the MinerU-on audit and uses the submitted MinerU-off grid. Relative to full RCPS, BGE-M3 alone changes only the near-tied 30B-Prod order ( $\tau = 0.80$ ); lower parser ranks and the chunker order remain unchanged. It provides no evidence about the MinerU-on-PaddleOCR order.

**Artifact status.** The repository contains the frozen 663-Q-A probe, MinerU-on outputs, retrieval and diagnostic code, audited training arrays, and aggregate parser and chunker grids. A deterministic script reproduces the 2,036-Q-A OHR compatibility mask and paired estimates. It does not yet contain MinerU-off outputs, final human adjudication labels, an aligned RADP-Distill artifact, complete run configurations and checkpoints, or per-Q-A arrays for the complete 294-page grid. A full OHR-Bench v2 rerun and clean-checkout verification also remain outstanding. We do not describe these items as released or reproduced.